Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad Cantonment, Natore-6431

Lesson Plan
First Year 1st semester
Course Title: Inorganic Quantitative Analysis
Course Instructor: Dr. Tasmina Khandaker
Lecturer, Department of Chemistry

Reference Books:

1. Inorganic Quantitative Analysis-Carroll Wardllaw Graffin. Palala press (2015)
2. Chemistry Raymond Chang
3. Analytical Chemistry and Quantitative Analysis- David S.Hage. James R Carr.Pearson.1st edition
4. Essentials of Physical Chemistry- B. S. Bahl and G. D. Tuli

Distribution of Marks:

Class Participation	5%
Observations	5%
Homework/ Assignment	5%
Class Test/Quiz	15%
Final Examination	70%
Total	100%

pH

When weak acids or weak bases are dissolved in water, they can contribute H^+ by ionizing or Consume H^+ by being Protonated. These Processes are also governed by equilibrium constants. The Total Hydrogen ion Concentration from all sources, which is experimentally measurable, is expressed as the pH of the solution.

Water molecules have a slight tendency to undergo reversible ionization to yield a hydrogen ion a hydroxide ion:

$$H_2O \rightleftharpoons H^+ + OH^-$$

- The Concept of pH was first introduced by the Danish chemist Sorenson in 1909. where, P means “puissance d” and H mean “Hydrogen”. It is the French word, which means strength/power of Hydrogen.

- **The formal definition of pH is the negative logarithm of the Hydrogen ion Activity.**

$$pH = -\log [H^+]$$

- pH is a unit of measure which describes the degree of acidity or alkalinity (basic) of a solution.
- It is measured on a scale of 0 to 14.
- If the H^+ concentration is higher then OH^-, the material is acidic and the pH is less than 7
- If the OH^- concentration is higher than H^+ the material is basic and the pH is more than 7
- Number of H^+ equals the number of OH^- 7 is neutral< is acidic > 7 is basic

Why pH measurements are important ?

1. Measurement of pH of a solution is necessary to judge whether the liquid is acidic , alkaline or neutral .
2. The application areas are food and beverage industry , aquarium , gardening, medicine, biology, chemistry, agriculture, forestry, food science , environmental science, oceanography, civil engineering and many other applications.
3. Measurement of pH of liquids is necessary in many condensate systems as a check on corrosion , in pharmaceuticals and drug production for maximum yield , in food manufacturing to reduce spoilage and improve taste and in innumerable chemical processes .
4. The continuous monitoring of blood pH is essential for proper treatment of patients suffering from metabolic and respiratory problems .

Measurement of pH:

The pH can be measured by:

(1) pH strips
(2) pH indicators
(3) pH meter

❖ **pH strips**: pH strips are pieces of paper that change color depending on the pH, the acidity or alkalinity- of a liquid.

- pH strips are Brownish yellow in colour (whatman filter); by this method approx. value is obtained, which are less accurate.

❖ pH indicators: Indicators are the organic Compounds of natural or Synthetic origin whose colour is dependent on pH.

-The pH indicators are either weak acids or weak bases having different colours in undissociated and dissociated forms:

$$\underset{\text{(colour-A)}}{HIn} \leftrightarrow H^+ + \underset{\text{(colour-B)}}{In^-}$$

- pH meter :-

-The pH meter is a laboratory equipment which used to measure acidity or alkalinity of a solution.

-The pH meter measures The concentration of Hydrogen ions [H⁺] using An ion-sensitive electrode .

-It is the most reliable and Convenient method for Measuring ph.

pH Scale:

In order to express the hydrogen ion concentration or acidity of a solution, a pH scale was evolved.

The pH is defined as pH = – log [H+] or

[H+] = 10– pH

The hydrogen ion concentration of different acidic solutions were determined experimentally. These were converted to pH values using the above relations.

Then these pH values were computed on a scale taking water as the reference substance. The scale on which pH values are computed is called the pH scale. Water dissociates to H+ and OH– ions to a very small degree so that we have the equilibrium.

Water dissociates to H^+ and OH^- ions to a very small degree so that we have the equilibrium.

$$H_2O \rightleftharpoons H^+ + OH^- \quad ...($$

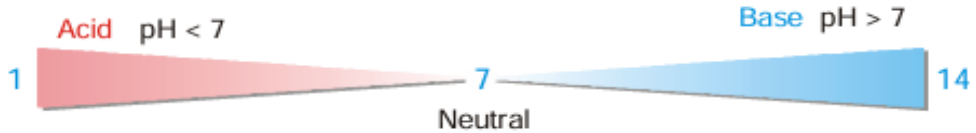
We can write the equilibrium expression as

$$K = \frac{[H^+]\,[OH^-]}{[H_2O]} \quad ...($$

Since water is so little dissociated, the concentration of undissociated molecules, $[H_2O]$, presumed to be constant. Therefore from expression (2) we can write

$$[H^+]\,[OH^-] = K\,[H_2O] = K_w = \text{a constant.} \quad ...($$

where Kw is called the water dissociation constant or the water ionisation constant.



How to calculate $[H^+]$ and $[OH^-]$ from K_w.

In any aqueous solution, the product of $[H^+]$ and $[OH^-]$ is always equal to K_w. This is so irrespective of the solute and relative concentrations of H^+ and OH^- ions. However, the value of K_w depends on temperature.

At 25°C it is 1.0×10^{-14}. Thus, $[H^+][OH^-] = 1.0 \times 10^{-14}$

Each of $[H^+]$ and $[OH^-]$ in pure water at 25°C is 10^{-7}. The concentration of H^+ and OH^- ions are expressed in gram moles per litre.

The concentration of H^+ and OH^- ions can be calculated from the expressions :

$$[H^+] = \frac{K_w}{[OH^-]}$$

$$[OH^-] = \frac{K_w}{[H^+]}$$

Relation between pH and pOH

We have already stated that pH concept can also be used to express small quantities as $[OH^-]$ and K_w.

Thus we have $pOH = -\log [OH^-]$

$$pK_w = -\log k_w$$

Let us consider the log form of the expression

$$K_w = [H^+][OH^-]$$

That is, $\log K_w = \log [H^+] + \log [OH^-]$

$$-\log K_w = -\log [H^+] - \log [OH^-]$$

$$pK_w = pH + pOH$$

$$K_w = 1.0 \times 10^{-14}$$

$$pKw = -\log (1.0 \times 10^{-14}) = 14.00$$

Hence, for any aqueous solution at 25°C, pH and pOH add up to 14.00. That is, pH + pOH = 14.00

PROBLEM 1. The hydrogen ion concentration of a fruit juice is 3.3×10^{-2} M. What is the pH of the juice ? Is it acidic or basic ?

SOLUTION:

The definition of pH is pH =– log [H+]

We are given[H+] = 3.3×10^{-2}

Substituting into the definition of pH,

we get pH =– log (3.3×10^{-2}) =– (– 1.48) = 1.48

Since the pH is less than 7.00 the solution is acidic.

PROBLEM 2. If a solution has a pH of 7.41, determine its H+ concentration.

SOLUTION:

Since pH =– log [H+]– pH = log [H+] Now [H+] can be calculated by taking antilog of – pH : [H+] = antilog (– pH)

In this case, [H+] = antilog (– 7.41) = $10^{-7.41}$

Therefore, the [H+] of the given solution is 3.9 × 10– 8 M

Buffer solution

A buffer solution is one which maintains its pH fairly constant even upon the addition of small amounts of acid or base.

There are two types of buffer solution.
Acidic Buffer
Basic Buffer

Acidic Buffer: A weak acid together with a salt of same acid with a strong base. These are called Acidic Buffer.

For Example.

$CH_3COOH + CH_3COONa$.

Basic Buffer: A weak base and its salt with a strong acid. These are called Basic Buffers.

For Example.

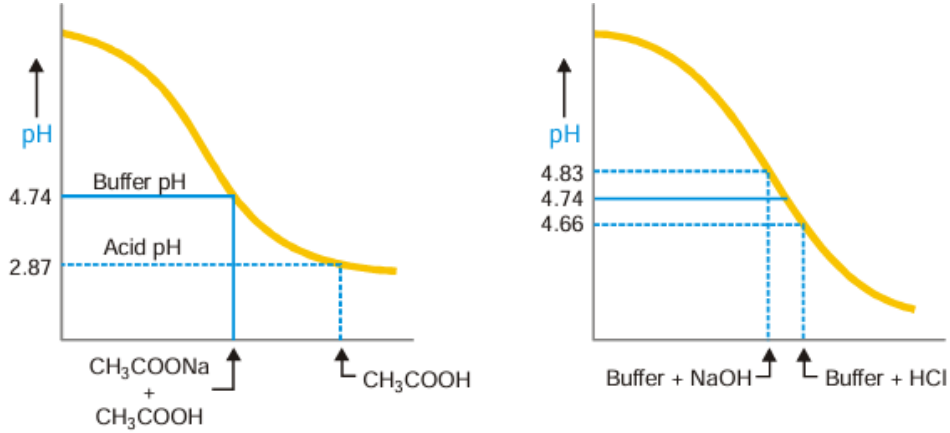
$NH_4OH + NH_4CL$

Let us illustrate buffer action by taking example of a common buffer system consisting of solution of acetic acid and sodium acetate (CH_3COOH/CH_3COONa).

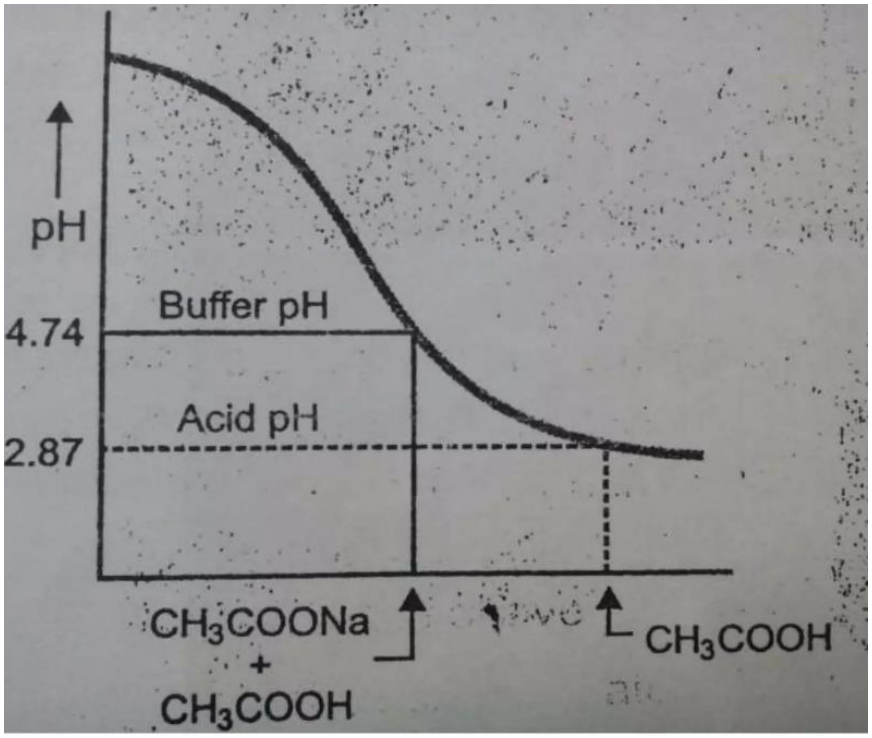
$$CH3COOH \rightleftharpoons H^+ + CH3COO^-$$

$$CH3COONa \longrightarrow Na^+ + CH3COO^-$$

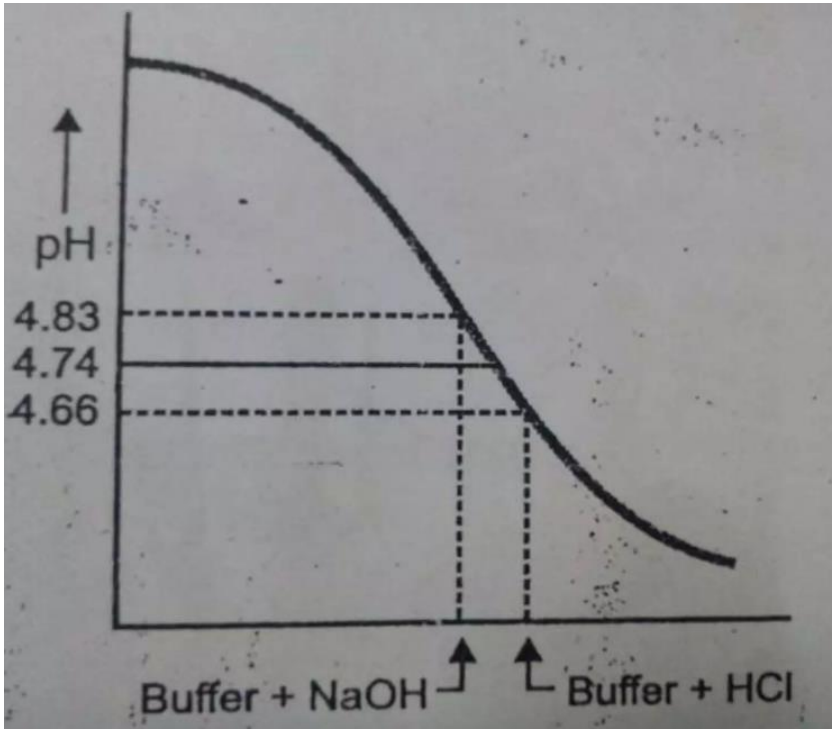
since the salt is completely ionised, it provides the common ions CH_3COO^- in excess. The common ion effect suppresses the ionisation of acetic acid. This reduces the concentration of H^+ ions which means that pH of the solution is raised. Thus, a 0.1 M acetic acid solution has a pH of 2.87 but a solution of 0.1 M acetic acid and 0.1 M sodium acetate has a pH of 4.74 (Fig.). Thus 4.74 is the pH of the buffer. On addition of 0.01 mole NaOH the pH changes from 4.74 to 4.83, while on the addition of 0.01 mole HCl the pH changes from 4.74 to 4.66. Obviously the buffer solution maintains fairly constant pH and the changes in pH could be described as marginal.



The Buffer solution has a higher pH than the acid itself:



The pH of Buffer changes only slightly upon addition of an acid or base



HOW A BUFFER OPERATES ?

We have already stated that a buffer solution containing equimolar amounts (0.10 M) of acetic acid and sodium acetate has pH 4.74. Now we proceed to discuss how the addition of a small amount of HCl or NaOH to the buffer solution affects its pH. The pH of the buffer is governed by the equilibrium

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

The buffer solution has a large excess of CH_3COO^- ions produced by complete ionisation of sodium acetate, $CH_3COONa \longrightarrow CH_3COO^- + Na^+$

(1) Addition of HCl. Upon the addition of HCl, the increase of H^+ ions is counteracted by association with the excess of acetate ions to form unionised CH_3COOH. Thus the added H^+ ions are neutralised and the pH of the buffer solution remains virtually unchanged. However owing to the increased concentration of CH_3COOH, the equilibrium (1) shifts slightly to the right to increase H^+ ions. This explains the marginal increase of pH of the buffer solution on addition of HCl

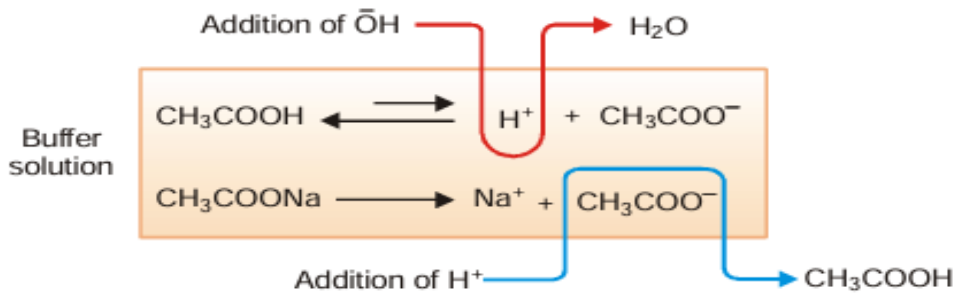


Figure 27.13
Mechanism of Buffer action of an acid buffer.

(2) Addition of NaOH. When NaOH is added to the buffer solution, the additional OH^- ions combine with H^+ ions of the buffer to form water molecules. As a result the equilibrium (1) shifts to the right to produce more and more H^+ ions till practically all the excess OH^- ions are neutralised and the original buffer pH restored. However, a new equilibrium system is set up in which [CH3COOH] is lower than it was in the original buffer. Consequently $[H^+]$ is also slightly less and pH slightly higher than the buffer pH values (Fig. 27.13). Operation of a Basic buffer as NH_4OH/NH_4Cl can also be explained on the same lines as of an acid buffer (Fig. 27.14) upon addition of HCl the H^+ ions combine with OH^- ions of the buffer to form water molecules.

The equilibrium,

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

is shifted to the right till all the additional H^+ ions are neutralised and the original buffer pH restored. When NaOH is added to the buffer solution, OH^- ions associate with excess of NH_4^+ ions to form unassociated NH_4OH. Thus the pH of the buffer is maintained approximately constant.

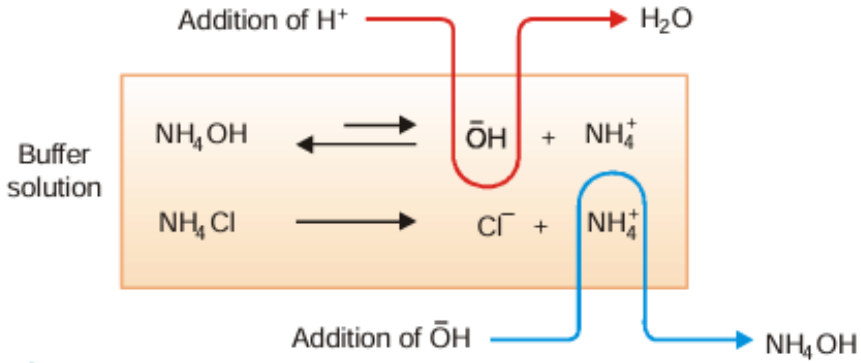


Figure 27.14
Mechanism of Buffer action of a basic buffer.

Calculate the pH of Buffer solutions:

The pH of an acid buffer can be calculated from the dissociation constant, Ka, of the weak acid and the concentrations of the acid and the salt used. The dissociation expression of the weak acid, HA, may be represented as

$$HA \rightleftharpoons H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$[H^+] = K_a \times \frac{[HA]}{[A^-]}$$

The weak acid is only slightly dissociated and its dissociation is further depressed by the addition of the salt (Na+A–) which provides A– ions (Common ion effect). As a result the equilibrium concentration of the unionised acid is nearly equal to the initial concentration of the acid. The equilibrium concentration [A–] is presumed to be equal to the initial concentration of the salt added since it is completely dissociated. Thus we can write the equation (1) as

$$[H^+] = K_a \times \frac{[\text{acid}]}{[\text{salt}]} \quad ...(2)$$

where [acid] is the initial concentration of the added acid and [salt] that of the salt used.

Taking negative logs of both sides of the equation (2), we have

$$-\log[H^+] = -\log K_a - \log\frac{[\text{acid}]}{[\text{salt}]} \quad ...(3)$$

But $\quad -\log[H^+] = pH \quad$ and $\quad \log K_a = pK_a$

Thus from (3) we have

$$pH = pK_a - \log\frac{[\text{acid}]}{[\text{salt}]} = pK_a + \log\frac{[\text{salt}]}{[\text{acid}]}$$

Hence $\quad \mathbf{pH = pK_a + \log\frac{[salt]}{[acid]}}$

This relationship is called the **Henderson-Hasselbalch equation** or simply **Henderson equation.**

In a similar way, the Henderson-Hasselbalch equation for a basic buffer can be derived. This can be stated as :

$$\mathbf{pOH = pK_b + \log\frac{[salt]}{[base]}}$$

Significance of the Henderson-Hasselbalch equation

With its help :

(1) The pH of a buffer solution can be calculated from the initial concentrations of the weak acid and the salt provided K_a is given.

However, the Henderson-Hasselbalch equation for a basic buffer will give pOH and its pH can be calculated as (14 – pOH).

(2) The dissociation constant of a weak acid (or weak base) can be determined by measuring the pH of a buffer solution containing equimolar concentrations of the acid (or base) and the salt.

$$\text{pH} = \text{p}K_a + \log\frac{[\text{salt}]}{[\text{acid}]}$$

Since $$[\text{salt}] = [\text{acid}], \quad \log\frac{[\text{salt}]}{[\text{acid}]} = \log 1 = 0$$

$\therefore$ $$\text{p}K_a = \text{pH}$$

The measured pH, therefore, gives the value of pK_a of the weak acid.

Likewise we can find the pK_b of a weak base by determining the pH of equimolar basic buffer.

(3) A buffer solution of desired pH can be prepared by adjusting the concentrations of the salt and the acid added for the buffer.

It is noteworthy that buffer solution are most effective when the concentrations of the weak acid (or weak base) and the salt are about equal. This means that pH is close to the value of pK_a of the acid (or pK_b of the base).

SOLVED PROBLEM 1. Find the pH of a buffer solution containing 0.20 mole per litre CH_3COONa and 0.15 mole per litre CH_3COOH. Ka for acetic acid is 1.8×10^{-5}

SOLVED PROBLEM 2. Estimate the pH at 25°C containing 0.10 M sodium acetate and 0.03 M acetic acid pKa for CH_3COOH = 4.57.

SOLVED PROBLEM 3. Calculate the pH of a buffer solution that is 0.250 M in formic acid, HCOOH, and 0.100 M in sodium formate, HCOONa. Ka for formic acid is 1.8×10^{-4}.

SOLVED PROBLEM 4. A buffer solution contains 0.015 mole of ammonium hydroxide and 0.025 mole of ammonium chloride. Calculate the pH value of the solution. Dissociation constant of NH_4OH at the room temperature is 1.80×10^{-5}